KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 16 · SCIENCE

Waves, Light and How We See

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Unit 3):** Name one energy change you learned about in Week 12, such as in a flashlight or a solar panel. Write the form of energy before and after.

2. Drop a pebble into a pond and rings of ripples spread out. What do you think is moving outward: the water itself, or something else? Explain your guess.

3. Draw a wave. Label where you think its highest point and lowest point are.

4. How do you think your eyes see a book on a table in a lit room? Draw arrows to show which way you think the light travels.

5. Name one way people send messages over long distances without talking face to face. How does it work?

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Show What I Learned

Use evidence from today's lesson. Write in complete sentences.

1. Draw a wave with a dashed rest line. Label the **crest**, **trough**, **amplitude** and **wavelength**.

2. **Two steps:** A wave drawing has crests at squares 3, 11 and 19, and each crest rises 4 squares above the rest line. (a) What is the wavelength? (b) What is the amplitude?

3. **Find the mistake:** "I can see my red shirt in a totally dark closet, because it is red." Explain what is wrong. Use the word **reflect**. Then name one thing that **refracts** light.

4. **Mirror Lab:** A beam hits a mirror at 25 degrees from the normal. Predict the reflected angle. Then explain why one trial that measured 60 and 58 degrees does not prove the law of reflection wrong.

★ **Challenge:** Use these Morse letters: A = · –, E = ·, H = · · · ·, I = · ·, N = – ·, O = – – –, S = · · ·, T = –. Write a 3-letter word in dots and dashes. Why are two kinds of signals hard to misread?

Looking Ahead: Next week: **Energy Resources.** Compare fossil fuels, nuclear, solar, wind, water, geothermal and biomass energy with real U.S. Energy Information Administration data. Bring this question: where does your electricity come from?